

Perfect Harshit — let's reset everything and I'll explain **exactly what your backend is doing, why each step is needed, and how to set it up again anytime**, even after restarting your laptop.

✅ **PART 1 — What Your Backend Is Actually Doing**

Your backend currently has **2 files**:

1 **gee_ndvi.py — Connects to Google Earth Engine + Generates NDVI**

This file:

✓ **Initializes Google Earth Engine**

Uses your service account & service-account key JSON.

```
ee.Initialize(project='epics-gee-ndvi-project')
```

(If using service account: ee.ServiceAccountCredentials... you already configured it.)

✓ **Converts your AOI (area polygon) from GeoJSON → EE Geometry**

```
aoi = ee.Geometry(aoi_geojson)
```

✓ **Loads Sentinel-2 satellite images**

```
ee.ImageCollection("COPERNICUS/S2_SR")
```

Filtered by:

- AOI
 - start date
 - end date
 - cloud percentage < 20%
-

✓ **Computes Median Image**

```
image = collection.median()
```

This reduces many satellite images into 1 clean image.

✓ Computes NDVI

```
ndvi = image.normalizedDifference(["B8", "B4"]).rename("NDVI")
```

(B8 = NIR, B4 = Red)

✓ Generates a thumbnail URL

```
url = ndvi.getThumbURL({...})
```

This **is what your frontend displays**.

2 app.py — Flask API

Your backend exposes **one API endpoint**:

POST /ndvi

It:

1. Accepts JSON: aoi + start + end
 2. Calls `get_ndvi()` from `gee_ndvi.py`
 3. Returns NDVI thumbnail URL to frontend
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✓ PART 2 — Why You Were Getting 404 Earlier

404 happens when:

✗ You opened:

`http://127.0.0.1:5000/ndvi`

in browser **without POST request** → browser only uses GET → Flask replies with 404.

POST only works from:

- ✓ Your frontend
- ✓ Postman
- ✓ cURL

NOT your browser URL bar.

✓ **PART 3 — IMPORTANT: Steps to Follow Every Time After RESTART**

Once a system restarts → Earth Engine logout happens → must re-authenticate

But you are using **service accounts**, so auth is automatic.

★ **FOLLOW THESE STEPS EXACTLY AFTER EVERY RESTART**

STEP 1 — Open your backend folder

```
cd D:\DESKTOP\ML\EPICS-PROJECT\backend
```

STEP 2 — Activate Virtual Environment

```
venv\Scripts\activate
```

You should see:

```
(venv)
```

STEP 3 — Make sure service-account key exists

Inside backend folder, file must exist:

```
backend/
```

```
  service-key.json
```

```
  app.py
```

```
  gee_ndvi.py
```

If you delete or move it → GEE will not initialize.

STEP 4 — Run Backend

```
python app.py
```

Server starts at:

```
http://127.0.0.1:5000
```

STEP 5 — Test API (Postman)

POST → URL:

http://127.0.0.1:5000/ndvi

Body (JSON):

```
{
  "aoi": {
    "type": "Polygon",
    "coordinates":
[[[76.7275,23.0275],[76.86,23.0275],[76.86,22.89],[76.7275,22.89],[76.7275,23.0275]]]
  },
  "start": "2023-01-01",
  "end": "2023-02-01"
}
```

You will get thumbnail URL.

✓ **PART 4 — How to Increase Thumbnail Image Size**

In **gee_ndvi.py** change:

```
url = ndvi.getThumbURL({
  "min": 0,
  "max": 1,
  "palette": ["red", "yellow", "green"],
  "region": aoi,
})
```

→ Increase size using **dimensions**:

```
url = ndvi.getThumbURL({
  "min": 0,
  "max": 1,
  "palette": ["red", "yellow", "green"],
  "region": aoi,
```

```
"dimensions": 1024 # 512, 1024, 2048 (bigger = clearer)
})
```

✅ **PART 5 — How to Get AOI (Polygon) Easily**

★ **Best way:**

Use Google Earth Engine Code Editor

<https://code.earthengine.google.com/>

Steps:

1. Click **Geometry Tool (square icon)**
 2. Draw polygon on map
 3. Right side panel → click "**Copy as GeoJSON**"
 4. Paste into your frontend AOI textarea
-

Alternative: Use geojson.io

<https://geojson.io>

- Draw polygon
 - Copy output GeoJSON
-

Alternative: Use QGIS

But for you, **geojson.io is easiest.**

🎉 **You're Now 100% Clean & Fully Working**

Your backend is correct.

Your errors were because of:

- ✓ No service account
- ✓ Wrong folder
- ✓ GET request instead of POST
- ✓ EE not initialized properly

Now everything is fixed.

If you want:

✅ I can rewrite your backend in a cleaner structure

or

✅ Connect frontend to backend fully

or

✅ Add NDWI, NDBI, EVI, time-series, charts

Just tell me ❤️